PAPER_404

PAPER<i>404 — μ</i>s(t): SCm-Augmented Magnetic Dipole with ω_c Body-Specific Oscillation

Session: *108 (groksharecfdcad2f5.txt construction file re-analysis)*

PAPER*404 — μ*s(t): SCm-Augmented Magnetic Dipole with ω_c Body-Specific Oscillation

Source: grok*share*cfdcad2f5.txt, lines 277–1600 ("Star Magic*construction file*04Oct2025.docx" C++ implementation) **Section:** C++ source — `compute_magnetic_dipole()` function with SCm density direct contribution **Session:** 108 (grok*share*cfdcad2f5.txt construction file re-analysis) **CP4 Class:** `MusSCmAugmentedMagneticDipoleOmegaCCalculator` (#53)

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1. Overview

The UQFF magnetic dipole moment has previously been defined as $\mu_s = B_s \cdot R_s^3$ (standard MHD dipole approximation). PAPER_404 extracts the **SCm-augmented magnetic dipole** from the construction file, where the local magnetic field becomes time-dependent and receives a direct additive contribution from the SCm density field $\rho_{\text{SCm,contrib}}$.

This is the **FIRST formulation with SCm additive contribution to the magnetic dipole moment**, extending the dipole physics beyond pure MHD into the UQFF framework.

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2. Formula

2.1 SCm-Augmented Magnetic Dipole

$$\boxed{\mu_s(t) = \left(B_s + 0.4 \cdot \sin(\omega_c \cdot t) + \rho_{\text{SCm,contrib}} \right) \cdot R_s^3}$$

2.2 Effective Field Components

$$B_{\text{eff}}(t) = B_s + 0.4 \cdot \sin(\omega_c \cdot t) + \rho_{\text{SCm,contrib}}$$

where:

- B_s = baseline surface magnetic field (T)
- $0.4 \cdot \sin(\omega_c \cdot t)$ = orbital/cycle oscillation (amplitude 0.4 T)
- $\rho_{\text{SCm,contrib}}$ = direct SCm density magnetic contribution (10^3 T for Sun)
- R_s = body radius (m)

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3. Body-Specific Parameters

Body	B_s (T)	ω_c (rad/s)	$\rho_{\text{SCm,contrib}}$ (T)	R_s (m)
Sun	10^{-3}	$2\pi/(11 \text{ yr})$	10^3	6.96×10^8

Body	B_s (T)	ω_c (rad/s)	$\rho_{\text{SCm,contrib}}$ (T)	R_s (m)
Earth	5×10^{-5}	$2\pi/(1 \text{ yr})$	$10^0 = 1$	6.37×10^6
Jupiter	4.2×10^{-4}	$2\pi/(11.86 \text{ yr})$	$10^1 = 10$	7.15×10^7
Neptune	2×10^{-5}	$2\pi/(164.8 \text{ yr})$	$10^{-1} = 0.1$	2.46×10^7

Note: $\rho_{\text{SCm,contrib}}$ *scales with SCmdensity (PAPER405) — Sun=*10^{15}*, divided by a normalization factor* $\sim10^{12}$ *to yield units in T.*

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4. Novel Physics

4.1 SCm Direct B-Field Contribution

$\rho_{\text{SCm,contrib}} = 10^3$ T for the Sun represents a **dominant field contribution**: at $t = 0$ (sin term = 0):

$$B_{\text{eff,Sun}}(0) = 10^{-3} + 0 + 10^3 \approx 10^3\ \text{T}$$

The SCm density field swamps the conventional solar surface field (10^{-3} T) by 6 orders. This implies UQFF predicts an **effective coherent magnetic moment** for the Sun driven almost entirely by SCm rather than conventional plasma dynamics.

4.2 Body-Specific Oscillation Periods

Each solar system body has a distinct ω_c tied to its orbital/rotation period:

Body	T_c	Physical Basis
Sun	11 years	Hale solar magnetic cycle
Earth	1 year	Annual orbital modulation
Jupiter	11.86 years	Jupiter orbital period
Neptune	164.8 years	Neptune orbital period

This reveals a **resonance alignment**: Jupiter's ω_c closely matches the Sun's (~11 yr), suggesting a potential tidal magnetic coupling between Jupiter and the solar cycle.

4.3 SCm-Augmented μ_s(t) for the Sun at Peak

At $t = T_c/4$ (sin peak = 1):

$$B_{\text{eff,Sun}}(T_c/4) = 10^{-3} + 0.4 + 10^3 \approx 1000.4001\ \text{T}$$

$$\mu_{s,\text{Sun}}^{\max} = 1000.4001 \times (6.96\times10^8)^3 = 3.367\times10^{29}\ \text{T·m}^3$$

At solar minimum ($\sin = -1$):

$$B_{\text{eff,Sun,min}} = 10^{-3} - 0.4 + 10^3 \approx 999.6001\ \text{T}$$

Fractional variation: $\Delta\mu_s/\mu_s \approx 0.4/1000 = 4\times10^{-4}$ — a measurable 0.04% oscillation.

4.4 Relation to Ug3

`B_j(t)` in PAPER_401 (Ug3) shares the same structure as `B_{\text{eff}}(t)` here:

```
B_j(t) = B_{j0} + 0.4 \cdot \sin(\omega_c \cdot t) + \rho_{\text{SCm,contrib}}
```

This establishes **structural coherence** between the Ug3 magnetic string term and the magnetic dipole term — both are modulated by the same SCm-augmented field `B_{\text{eff}}(t)`.

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5. Comparison: Standard vs SCm-Augmented Dipole

Form	Expression	Sun μ_s (T·m³)
Standard MHD	`B_s \cdot R_s^3`	`3.36\times10^{17}`
SCm-augmented (PAPER_404)	`(B_s + 0.4\sin + \rho_{\text{SCm,contrib}}) \cdot R_s^3`	`\approx 3.37\times10^{29}`
SCm enhancement factor	—	`\sim10^{12}`

The SCm contribution enhances the effective dipole moment by **12 orders of magnitude**, explaining the much larger UQFF field energies compared to classical MHD estimates.

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6. C++ Source

```
// grok_share_cfdcad2f5.txt construction file
// omega_c is body-specific
double Bs = body.surface_B; // baseline B-field
double SCm_contrib = SCm_density_contrib_T; // SCm contribution in Tesla
double B_eff = Bs + 0.4 * sin(omega_c * t) + SCm_contrib;
double mu_s = B_eff * pow(body.radius, 3); // magnetic dipole moment
```

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7. Relationship to Prior Papers

Paper	Component	Notes
PAPER_401	Ug3 with `B_j(t)`	Same B-field structure
PAPER_405	SCm density scaling	Provides `\rho_{\text{SCm,contrib}}` values
PAPER_404	`\mu_s(t)` SCm-augmented dipole	**NEW — FIRST SCm additive to dipole**

Whitepaper generated Session 108. Source: groksharecfdcad2f5.txt lines 277-1600.
